

Production Deployment Guide - FAQ Chatbot (Qdrant Version)

Follow this step-by-step guide to deploy the FastAPI FAQ Chatbot backend and connect it to your production Qdrant instance.

Step 1: Install Server-Level Prerequisites

Log in to your production server terminal and install Python, PostgreSQL, and Docker (if needed):

```
# 1. Update the server package repository
sudo apt update && sudo apt upgrade -y

# 2. Install Python 3.10+, pip, and virtual environment utilities
sudo apt install python3 python3-pip python3-venv -y

# 3. Install PostgreSQL database server (for transactional users/sessions data)
sudo apt install postgresql postgresql-contrib -y

# 4. (Optional) Install Docker - ONLY if Qdrant is not already running on the server
# sudo apt install docker.io -y
# sudo systemctl enable --now docker
```

Step 2: Spin Up Qdrant (Skip if Qdrant is already running)

If a Qdrant service is already running on your production infrastructure, skip this step.

If you need to start a fresh instance, run the persistent container:

```
docker run -d \
  --name qdrant \
  --restart always \
  -p 6333:6333 \
  -p 6334:6334 \
  -v qdrant_storage:/qdrant/storage \
  qdrant/qdrant
```

Step 3: Upload Project Files

Copy your project code to the target directory on your server (e.g., `/var/www/faq_chatbot/`).

Keep these files/directories:

- `app/` (all agents, routes, data, and database logic)
- `migrations/` (SQL migration files for Postgres tables)
- `main.py`
- `run_migrations.py`
- `requirements.txt`
- `.env`

Exclude these (do not copy):

- `.venv/`
 - `__pycache__/`
 - `logs/`
-

Step 4: Set Up Python Virtual Environment

Navigate to the project directory on your server and initialize the Python environment:

```
# 1. Create a clean virtual environment
python3 -m venv venv

# 2. Activate the virtual environment
source venv/bin/activate

# 3. Upgrade pip and install all required libraries
pip install -U pip
pip install -r requirements.txt
```

Step 5: Configure the `.env` File

Open the `.env` file on the server and update it. Point the chatbot to your existing Qdrant service:

- Database Connection: Set `DATABASE_URL` and `CHECKPOINT_DB_URL` to point to your PostgreSQL server.
- Qdrant Connection:
 - * Set `QDRANT_URL` to your production Qdrant IP/domain (e.g., `QDRANT_URL=http://192.168.1.50:6333` or `http://localhost:6333`).
 - * If your production Qdrant requires an API key, set `QDRANT_API_KEY=your-production-api-key`. Otherwise, leave it empty.
 - * Set `QDRANT_COLLECTION=faq_embeddings`.
- LLM Provider: Configure the `LLM_PROVIDER` (e.g., `groq`, `openai`, or `gemini`) and set the corresponding API

keys.

- Urls: Update `FRONTEND\_URL` and `BACKEND\_URL` with your public production domains.
-

Step 6: Run Database Migrations (PostgreSQL)

Run the PostgreSQL database migration script to construct all the relational tables (`users`, `sessions`, `messages`, `logs`, etc.):

```
python run_migrations.py
```

Step 7: Verify and Run the Application

Start the FastAPI server using Uvicorn. On the first startup, the application lifespan will:

1. Connect to PostgreSQL and initialize the LangGraph checkpointer tables.
2. Connect to your active Qdrant database, create the `faq\_embeddings` collection, and automatically seed the FAQ questions and answers.
3. Download the semantic embedding model (if configured for online download).

To launch the server with multiple parallel workers for production traffic:

```
uvicorn main:app --host 0.0.0.0 --port 8000 --workers 4
```

Step 8: Manage Server Lifespan with Systemd

To keep the application running continuously in the background and ensure it automatically restarts if the server reboots or crashes, create a systemd service:

1. Create the service file:

```
sudo nano /etc/systemd/system/faq-chatbot.service
```

2. Paste the following configuration (replace path placeholders with your actual directory path):

```
[Unit]
```

```
Description=FastAPI FAQ Chatbot Service
```

```
After=network.target
```

```
[Service]
```

```
User=www-data
```

```
WorkingDirectory=/var/www/faq_chatbot
```

```
ExecStart=/var/www/faq_chatbot/venv/bin/uvicorn main:app --host 127.0.0.1 --port 8000
```

```
--workers 4
Restart=always

[Install]
WantedBy=multi-user.target
```

3. Enable and start the background service:

```
sudo systemctl daemon-reload
sudo systemctl enable faq-chatbot
sudo systemctl start faq-chatbot
```

Step 9: Configure Nginx as a Reverse Proxy with SSL (HTTPS)

Serve the chatbot backend securely over HTTPS:

1. Install Nginx and Certbot:

```
sudo apt install nginx certbot python3-certbot-nginx -y
```

2. Open Nginx configuration:

```
sudo nano /etc/nginx/sites-available/faq_chatbot
```

3. Add the server proxy block:

```
server {
    server_name yourdomain.com;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

4. Enable the site and restart Nginx:

```
sudo ln -s /etc/nginx/sites-available/faq_chatbot /etc/nginx/sites-enabled/
sudo systemctl restart nginx
```

5. Request and install an SSL certificate:

```
sudo certbot --nginx -d yourdomain.com
```